

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 1

Duration :60 Mins
On Class
Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Tamil Elakiya S (192524443) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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Technical Presentation

Language Processors: Compiler vs Interpreter

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Introduction to Language Processors

- A **language processor** converts a high-level programming language into machine code that a computer can execute.
- Since computers understand only binary instructions, language processors act as a bridge between programmers and hardware.
- The two major types of language processors are:
 - **Compiler**
 - **Interpreter**

Key Pointer :

"Without language processors, high-level programming languages like C, Java, and Python cannot be executed."



What is a Compiler?

Definition

A **compiler** translates the **entire source code** into machine code before execution.

Working Process

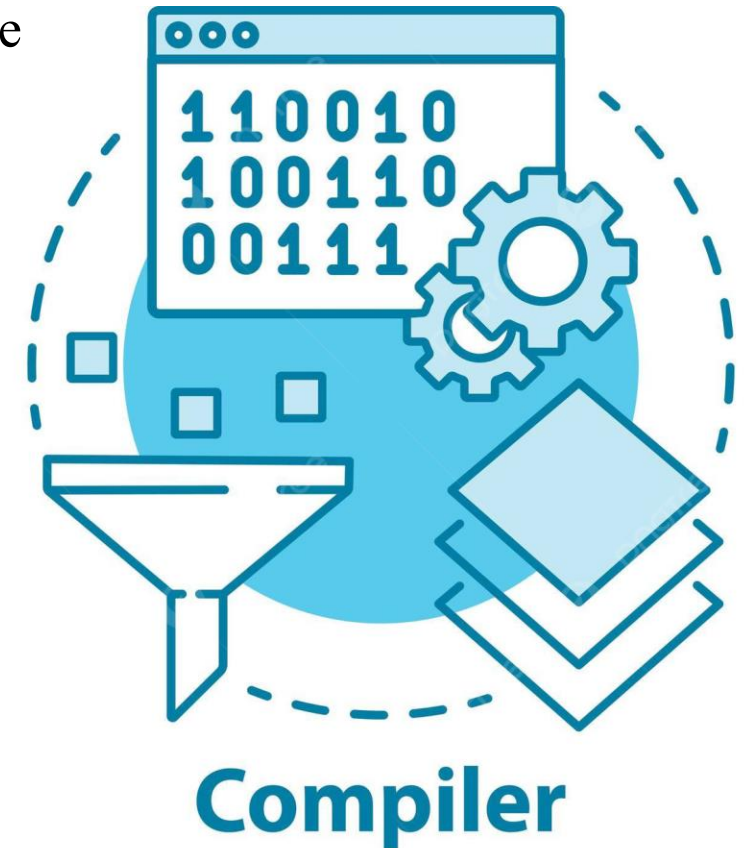
Source Code → Compiler → Object Code → Executable Program

Characteristics

- Translates the whole program at once.
- Generates an executable (.exe) file.
- Reports all errors after compilation.
- Faster execution after successful compilation.
- Requires recompilation after code modification.

Examples

- C ,C++ ,Go , Rust



What is an Interpreter?

Definition

An interpreter translates and executes the program one statement at a time.

Working Process

Source Code → Interpreter → Execute Line by Line

Characteristics

- No executable file is created.
- Stops immediately when an error occurs.
- Easier debugging.
- Slower execution because translation happens during runtime.

Examples

- Python , JavaScript , Ruby , PHP



Interpreter

Compiler vs Interpreter

Feature	Compiler	Interpreter
Translation	Entire program	One statement at a time
Execution Speed	Fast	Slower
Error Detection	After compilation	During execution
Executable File	Generated	Not generated
Memory Usage	More	Less
Debugging	Difficult	Easy
Program Execution	After compilation	Immediate

Real-World Applications

Compiler Applications

- Operating Systems
- Device Drivers
- Game Engines
- Embedded Systems
- Desktop Applications

Examples:

- Windows software
- Android native apps
- Video games

Interpreter Applications

- Web Development
- Data Science
- Artificial Intelligence
- Automation Scripts
- Web Browsers

Examples:

- Python
- JavaScript
- Machine Learning

Conclusion

- Both compiler and interpreter convert high-level code into machine-understandable instructions.
- Compilers provide faster execution and better optimization, making them suitable for performance-critical software.
- Interpreters simplify development by executing code line by line and providing immediate error feedback.
- The choice depends on the application's requirements, such as execution speed, portability, and ease of debugging.





Geo-Tag Photo

The image features a light blue background adorned with various autumn-themed elements. In the top left corner, there is a large, vibrant red maple leaf. To its right, a small cluster of blueberries and a single walnut are visible. Further right, a delicate branch with small, dried brown leaves is positioned. In the top right corner, a large, bright yellow maple leaf is displayed, accompanied by more blueberries. The bottom left corner features a large yellow maple leaf and a small group of blueberries. Along the bottom edge, there are several more elements: a single blueberry, a small branch with dried leaves, a walnut, a small cluster of blueberries, and a few more autumn leaves in shades of red and orange. The text "thank you" is centered in a dark blue, elegant script font.

thank you